

Ordered Quaternionic Response Quotients

Flat Cores, Local Residuals, and a Three-Spectator Descent Obstruction

Residual Chart Lab

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Abstract

We study finite cokernels formed from four ordered quaternion-valued face responses and their six pairwise common shadows. The incidence of labelled edge images distinguishes quotients with the same abstract dimension. The one-spectator family admits two explicit coordinate charts with a central cross-product transition. At two internal spectators, six reduced spacing words carry a flat 144-dimensional stationary-edge core connection, while the word 212 has an additional quaternionic residual invisible to its incident hinges. We then construct the three-spectator quotient at 222. It has dimension 444, an intrinsic 432-dimensional outer-edge subspace, and a 12-dimensional residual quotient. A specified central-slot right decoder induces an isomorphism from the 212 residual tensored with the vector representation onto this new residual quotient. The same suspension fails to descend to the complete quotients: its obstruction image is exactly the 432-dimensional core. All finite rank claims are over the rationals and are accompanied by exact arithmetic certificates. The comparison operator is part of the theorem; no choice-independent full transport or nonzero core curvature is asserted.

Snapshot scope. This research snapshot consolidates the results for mathematical review. The immutable Free Numbers Core v1.0.0 artifact is a cited foundation, not a version replaced by this snapshot. The manuscript reorganizes established source notes and the Note 26 result; it does not claim a complete three-spectator atlas or a physical interpretation. This is revision research-2026-09-07-audit1; the companion audit report records the corrections and the verification boundary.

Overview

We first describe the finite atlas through two internal spectators, then construct the residual comparison for the three-spectator word 222.

Four ordered face responses meet along six labelled edge responses. At the minimal tetrahedral level their compatibility quotient is

$$\mathbb{H} \otimes \mathbb{H} = W_{12} \overset{\perp}{\oplus} K_4, \quad \dim W_{12} = 12, \quad K_4 \cong \mathbb{H}. \quad (0.1)$$

Insert one spectator among the five ordered gap positions. Each of the five resulting local quotients still has the same abstract 48-dimensional type

$$T_6 = (\mathbb{H} \otimes \mathbb{H}) \otimes V, \quad V = \text{Im } \mathbb{H}. \quad (0.2)$$

Nevertheless, no single direct seed coordinate covers all five placements. Two chart families are necessary:

$$U_L = \{1, 2, 3\}, \quad U_R = \{1, 3, 4, 5\}. \quad (0.3)$$

Their exterior overlap is trivial, while their central overlap is not:

$$G_1 = I, \quad G_3 = \text{id}_H \otimes \theta. \quad (0.4)$$

For $a, w \in V$,

$$\boxed{\begin{aligned} \theta(1 \otimes w) &= 1 \otimes w + \sum_{\rho=1}^3 e_\rho \otimes (e_\rho \times w), \\ \theta(a \otimes w) &= -w \otimes a, \end{aligned}} \quad a \times w = \frac{1}{2}(aw - wa). \quad (0.5)$$

Thus the abstract local module does not record the whole object. The incidence of its labelled edge images and the transition between valid local coordinates retain where the spectator was inserted.

The shortest accurate summary is therefore

$$\boxed{\text{same local quotient type} \not\equiv \text{same placement coordinate},} \quad (0.6)$$

and the first nontrivial coordinate change is an explicit cross-product shear.

1. Minimal setup

Let

$$\mathbb{H} = \text{span}_{\mathbb{R}}\{1, i, j, k\}, \quad V = \text{Im } \mathbb{H}. \quad (1.1)$$

For response depth m , put

$$\mathcal{R}_m = \text{Hom}(V^{\otimes m}, \mathbb{H}). \quad (1.2)$$

Choose four ordered gaps

$$Q = \{q_1 < q_2 < q_3 < q_4\}. \quad (1.3)$$

They are treated as the four vertices of an ordered tetrahedron. The six local edges are labelled

$$12, 13, 14, 23, 24, 34. \quad (1.4)$$

Four face responses restrict to the six pairwise common shadows. Their signed disagreement defines the matching map

$$\partial_{n,Q} : \mathcal{R}_{n-2}^{\oplus 4} \longrightarrow \mathcal{R}_{n-3}^{\oplus 6} \quad (1.5)$$

$$(\partial_{n,Q} F)_{ab} = \rho_{q_a}^{q_a q_b} F_a - \rho_{q_b}^{q_a q_b} F_b. \quad (1.6)$$

The local relation object is the cokernel

$$Y_{n,Q} := \text{coker } \partial_{n,Q}. \quad (1.7)$$

For each labelled edge, retain its image in the quotient:

$$E_{ab}^{(n,Q)} := \text{im} \left(\mathcal{R}_{n-3}^{(ab)} \longrightarrow Y_{n,Q} \right). \quad (1.8)$$

The restrictions in (1.6) have the following concrete construction. For $T = v_1 \otimes \cdots \otimes v_n$ and a set of omitted gaps $S \subset \{1, \dots, n-1\}$, set

$$A_{n,S}(T)(x) = v_n b_{n-1} v_{n-1} \cdots b_1 v_1, \quad b_g = \begin{cases} 1, & g \in S, \\ x_g, & g \notin S. \end{cases}$$

Every terminal face $A_{n,\{r\}}$ is onto. One can see this by replacing $v_{r+1}v_r$ by an arbitrary quaternion h , then successively using the invertible one-variable left and right encoders on the remaining probes. The right encoder is

$$\Theta_R(h \otimes w)(z) = hzw, \quad J_m(h; v_1, \dots, v_m)(x_1, \dots, x_m) = hx_m v_m \cdots x_1 v_1.$$

The left encoder is $\Theta_L(w \otimes h)(z) = wzh$. Their matrices in the quaternion basis are invertible. Thus the common-shadow restriction, when factored through a face, is unique:

$$\rho_r^{rs} A_{n,\{r\}} = A_{n,\{r,s\}}.$$

Its existence is the local decoder-collapse identity of Note 12. For all finite supports reported here it is also checked entry by entry against the literal product above. This fixes the matching map without choosing a basis of its cokernel. It does not require the global all-length filling theorem.

We work over the rational quaternion algebra with $i^2 = j^2 = k^2 = ijk = -1$, and extend scalars to $\mathbb{R}$ for the usual diagonal $SO(3)$ -action. All reported rational ranks therefore also hold over $\mathbb{R}$. The word *atlas* refers here to a discrete family of quotient coordinates and their labelled comparison maps; no ambient manifold is assumed. Unless actual gap labels are explicitly stated, edge labels ab denote the a -th and b -th support vertices. Local 14, for example, is the long edge even when its actual gap labels are $(1, 6)$.

The object relevant to placement is not only $Y_{n,Q}$, but the decorated quotient

$$\mathcal{Y}_{n,Q} = (Y_{n,Q}; E_{12}, E_{13}, E_{14}, E_{23}, E_{24}, E_{34}). \quad (1.9)$$

This distinction is the starting point of the atlas.

2. The seed tetrahedron at $n = 5$

At $n = 5$, there is one four-gap support. Note 14 replaces an abstract quotient basis by a closed, $SO(3)$ -equivariant operator

$$\omega_5 : \mathcal{R}_2^{\oplus 6} \longrightarrow \mathbb{H} \otimes \mathbb{H} \quad (2.1)$$

satisfying

$$\boxed{\omega_5 \partial_5 = 0, \quad \text{rank } \omega_5 = 16, \quad \ker \omega_5 = \text{im } \partial_5.} \quad (2.2)$$

Hence

$$Y_{5,G_5} \cong \mathbb{H} \otimes \mathbb{H}. \quad (2.3)$$

Put $e_0 = 1, e_1 = i, e_2 = j, e_3 = k$. Define five maps

$$P, A_0, B_0, C_0, D_0 : \mathbb{H} \otimes V \otimes V \longrightarrow \mathbb{H} \otimes \mathbb{H}$$

by

$$\begin{aligned} P(h, u, v) &:= h \otimes uv, \\ A_0(h, u, v) &:= \sum_{a=0}^3 he_a \otimes uve_a, \\ B_0(h, u, v) &:= \sum_{a=0}^3 he_a \otimes ue_a v, \\ C_0(h, u, v) &:= \sum_{a=0}^3 e_a \otimes uvhe_a, \\ D_0(h, u, v) &:= \sum_{a=0}^3 e_a he_a v \otimes u. \end{aligned}$$

Associativity fixes all parentheses. Each map is $SO(3)$ -equivariant: quaternion multiplication is equivariant for conjugation, and every summed pair of basis elements comes from the invariant tensor $\sum_{a=0}^3 e_a \otimes e_a$.

The six ordered blocks are

$$\begin{aligned} \Lambda_{12} &= P - \frac{1}{2}B_0 + \frac{1}{2}C_0 - D_0, \\ \Lambda_{13} &= P - \frac{1}{4}A_0 - \frac{1}{2}B_0 + \frac{1}{2}C_0 - D_0, \\ \Lambda_{14} &= -\frac{1}{4}A_0 - \frac{1}{2}B_0, \\ \Lambda_{23} &= -\frac{1}{4}A_0 - \frac{1}{2}C_0, \\ \Lambda_{24} &= P - \frac{1}{4}A_0, \quad \Lambda_{34} = P. \end{aligned}$$

Thus $\omega_5(F) = \sum_{a<b} \Lambda_{ab} J_2^{-1} F_{ab}$. Substitution gives four signed vertex identities

$$\sum_{b>a} \Lambda_{ab} J_2^{-1} \rho_{q_a}^{q_a q_b} = \sum_{b<a} \Lambda_{ba} J_2^{-1} \rho_{q_a}^{q_b q_a},$$

which prove $\omega_5 \partial_5 = 0$. The block $\Lambda_{34} = P$ is onto, and the matching matrix has rank 200 in its 216-dimensional target. These finite identities and ranks are verified over exact rational arithmetic by the seed certificate. They prove (2.2).

Write Λ_{ab} for the six edge blocks of ω_5 . Their ranks are

$$\begin{array}{c|cccccc} ab & 12 & 13 & 14 & 23 & 24 & 34 \\ \hline \text{rank } \Lambda_{ab} & 16 & 12 & 16 & 4 & 12 & 16. \end{array} \quad (2.4)$$

The two rank-12 images coincide, while the central rank-4 image supplies a complement:

$$W_{12} := \text{im } \Lambda_{13} = \text{im } \Lambda_{24}, \quad K_4 := \text{im } \Lambda_{23}, \quad (2.5)$$

$$\mathbb{H} \otimes \mathbb{H} = W_{12} \oplus K_4. \quad (2.6)$$

Note 18 makes this split canonical. Define

$$\iota(q) = \sum_{\alpha=0}^3 e_\alpha \otimes q e_\alpha, \quad \nu = \frac{1}{4} \iota^*, \quad (2.7)$$

where $e_0 = 1, e_1 = i, e_2 = j, e_3 = k$. Then

$$\boxed{K_4 = \iota(\mathbb{H}), \quad W_{12} = \ker \nu = \iota(\mathbb{H})^\perp.} \quad (2.8)$$

Thus the seed contains a canonical orthogonal $12 + 4$ channel split, not merely a dimension decomposition.

The adjoint uses the Euclidean quaternion inner product and its tensor product, with $1, i, j, k$ orthonormal. Explicitly, $\iota^*(x \otimes y) = y\bar{x}$ and $\iota^* \iota = 4I$.

The quaternion coordinate of the six edge blocks has the exact incidence pattern

ab	12	13	14	23	24	34
$\text{rank}(\nu \Lambda_{ab})$	4	0	4	4	0	4
outer sign	+	0	-	central	0	+

(2.9)

This labelled pattern is the seed datum to be compared after a spectator is inserted.

3. One spectator and five placements

Let

$$B_5 = \{1, 2, 3, 4, 5\}, \quad Q_s = B_5 \setminus \{s\}. \quad (3.1)$$

The omitted gap s is the spectator position. For every s ,

$$\dim Y_{6, Q_s} = 48, \quad Y_{6, Q_s} \cong T_6 := (\mathbb{H} \otimes \mathbb{H}) \otimes V. \quad (3.2)$$

The equality of abstract type does not produce one common seed coordinate. Use the three full-rank seed edges as possible anchors:

$$L = 12, \quad M = 14, \quad R = 34. \quad (3.3)$$

Exact rational classification gives

s	Q_s	L	M	R
1	(2, 3, 4, 5)	✓	✓	✓
2	(1, 3, 4, 5)	✓	✓	—
3	(1, 2, 4, 5)	✓	—	✓
4	(1, 2, 3, 5)	—	—	✓
5	(1, 2, 3, 4)	—	—	✓.

(3.4)

Every marked entry determines a unique surjective quotient coordinate

$$\Omega_s^e : (\mathbb{H} \otimes V^{\otimes 3})^{\oplus 6} \longrightarrow T_6 \quad (3.5)$$

whose kernel is exactly the image of the local matching map:

$$\ker \Omega_s^e = \text{im } \widehat{\partial}_{6, Q_s}. \quad (3.6)$$

The unmarked entries are impossible over $\mathbb{Q}$, not merely absent from one computation.

For precision, the direct normalization on the anchor edge e is

$$(\Omega_s^e)_e(h; v_1, v_2, v_3) = \Lambda_e(h; v_{a_1}, v_{a_2}) \otimes v_t.$$

Here the three remaining probe slots are in increasing gap order; $a_1 < a_2$ are the two slots belonging to the support and t is the spectator slot. Compatibility uniquely completes each marked anchor.

The middle chart adds no new family. The two essential domains are

$$U_L = \{1, 2, 3\}, \quad U_R = \{1, 3, 4, 5\}, \quad U_L \cup U_R = B_5. \quad (3.7)$$

This is the first exact placement atlas in the program.

4. The overlap transition

The two charts overlap at one exterior placement and one central placement:

$$U_L \cap U_R = \{1, 3\}. \quad (4.1)$$

At $s = 1$,

$$\Omega_1^L = \Omega_1^R. \quad (4.2)$$

At $s = 3$,

$$\Omega_3^L = G\Omega_3^R, \quad G = \text{id}_{\mathbb{H}} \otimes \theta. \quad (4.3)$$

Put

$$\delta(w) = \sum_{\rho=1}^3 e_\rho \otimes (e_\rho \times w). \quad (4.4)$$

Then the transition is the closed map

$$\begin{aligned} \theta(1 \otimes w) &= 1 \otimes w + \delta(w), \\ \theta(a \otimes w) &= -w \otimes a \quad (a, w \in V). \end{aligned} \quad (4.5)$$

The first line contains the essential alternating correction. A spectator outside the active region gives no correction; a spectator between the two active pairs forces a cross-product term.

The internal decomposition

$$\mathbb{H} \otimes V = (1 \otimes V) \oplus \delta(V) \oplus \text{Sym}^2 V \quad (4.6)$$

puts θ into the form

$$\theta = R + N, \quad R^2 = I, \quad N^2 = 0, \quad \text{rank } N = 3. \quad (4.7)$$

Its minimal polynomial is

$$\boxed{m_\theta(t) = (t - 1)^2(t + 1).} \quad (4.8)$$

Therefore the transition is not a permutation of tensor slots. It is an involutive reflection together with a three-dimensional nilpotent shear. Over characteristic zero it has infinite order.

In (4.7), R is the identity on $1 \otimes V$ and minus the tensor flip on $V \otimes V$; $N(1 \otimes w) = \delta(w)$ and $N(V \otimes V) = 0$. In particular $RN = NR$.

This is a statement about a coordinate transition. No repeated power in (4.8) is yet interpreted as a geometric loop.

5. What the placement remembers

At the central support

$$Q_3 = (1, 2, 4, 5), \quad (5.1)$$

the long edge $14 = (1, 5)$ has image dimension 44 rather than 48. Its four-dimensional quotient is detected by the cap residual of Note 17.

In the right chart define

$$\beta_R((x \otimes y) \otimes w) = xw\bar{y}. \quad (5.2)$$

On actual responses define

$$\lambda_3^L(F) = \sum_{a,b=1}^3 e_b F(e_a, e_a, e_b), \quad \lambda_3^R(F) = \sum_{a,b=1}^3 F(e_b, e_a, e_a) e_b.$$

The chart in (3.5) uses decoded edge tensors, so write $\hat{\lambda}_3^\bullet = \lambda_3^\bullet J_3$ and $\hat{\chi}_6 = \chi_6(\oplus_e J_3)$. This makes the domains in the following identity explicit.

Then

$$\boxed{\beta_R \Omega_3^R = \hat{\chi}_6 = (-\hat{\lambda}_3^L, +\hat{\lambda}_3^L, 0, 0, -\hat{\lambda}_3^R, +\hat{\lambda}_3^R).} \quad (5.3)$$

Identify Y_{6,Q_3} with T_6 by the induced right chart, so that the edge images below are subspaces of T_6 . Then

$$\boxed{E_{14} = \ker \beta_R, \quad T_6/E_{14} \cong \mathbb{H}.} \quad (5.4)$$

Tensor the seed split (2.8) with V :

$$T_6 = (W_{12} \otimes V) \overset{\perp}{\oplus} (K_4 \otimes V). \quad (5.5)$$

The direct extension of the seed quaternion does not survive the long-edge quotient:

$$\boxed{\beta_R(K_4 \otimes V) = 0, \quad K_4 \otimes V \subset E_{14}.} \quad (5.6)$$

The surviving quaternion is instead a quotient of $W_{12} \otimes V$. Thus the spectator does not carry the seed channel forward by a placement-blind tensor product. It changes which channel represents the residual.

The transition G explains this transfer. It mixes the two summands in (5.5), and in particular

$$G(K_4 \otimes V) \cap (K_4 \otimes V) = 0. \quad (5.7)$$

The seed quaternion has not been destroyed. Its coordinate realization has changed.

6. The residual is independent of the chart

Transporting β_R across the central transition gives

$$\beta_L := \beta_R G^{-1}, \quad (6.1)$$

with the closed formula

$$\beta_L((x \otimes y) \otimes w) = x(2y + \bar{y})w. \quad (6.2)$$

The same observable is recovered from either chart:

$$\boxed{\hat{\chi}_6 = \beta_R \Omega_3^R = \beta_L \Omega_3^L.} \quad (6.3)$$

This separates two roles:

- G records how the local coordinate changes with placement;
- χ_6 is the quaternionic residual unchanged by that coordinate change.

The coefficient $2y + \bar{y}$ is not new bookkeeping. The same scalar-vector weighting already occurs in the canonical central coordinate of the $n = 5$ seed. The atlas transition forces it to reappear.

7. The exceptional two-spectator quotient at $n = 7$

The $n = 6$ atlas is complete for one spectator. Notes 16 and 21 identify and then fully coordinate the exceptional two-spectator continuation.

The first ordered support with parity separation

$$OO \mid EE \quad (7.1)$$

is uniquely

$$\boxed{(1, 3 \mid 4, 6),} \quad (7.2)$$

whose spacing is $(2, 1, 2)$. On the four cross edges define

$$\kappa_{212}(F_{13}, F_{14}, F_{23}, F_{24}) = \varepsilon_4(F_{13}) - \varepsilon_4(F_{14}) - \varepsilon_4(F_{23}) + \varepsilon_4(F_{24}), \quad (7.3)$$

where

$$\varepsilon_4(F) = \sum_{a,b=1}^3 F(e_a, e_a, e_b, e_b). \quad (7.4)$$

Then, over $\mathbb{Q}$,

$$\boxed{\mathcal{R}_5^{\oplus 4} \xrightarrow{\partial_{\square}} \mathcal{R}_4^{\oplus 4} \xrightarrow{\kappa_{212}} \mathbb{H} \longrightarrow 0, \quad \ker \kappa_{212} = \text{im } \partial_{\square}.} \quad (7.5)$$

Thus the exceptional four-dimensional residual at $n = 7$ is the complete cokernel of an alternating quaternionic cross square. Its position is forced by order and parity, not guessed from the dimension anomaly.

Note 21 transports a full-rank edge anchor from the $n = 6$ atlas and proves that it has a unique compatible completion

$$C_{212} : E_{(1,3,4,6)} \longrightarrow T_7 := (\mathbb{H} \otimes \mathbb{H}) \otimes V^{\otimes 2}. \quad (7.6)$$

The core coordinate and the square residual exactly exhaust the quotient:

$$\boxed{Y_{7,(1,3,4,6)} \cong T_7 \oplus \mathbb{H}, \quad \dim Y_{7,(1,3,4,6)} = 144 + 4 = 148.} \quad (7.7)$$

The six labelled edge ranks split as

	12	13	14	23	24	34	
core	144	108	144	36	108	144	
full exceptional	144	112	148	40	112	144.	

(7.8)

Thus the extra quaternion is transverse to the transported 144-dimensional spectator core and is seen exactly on the four cross edges.

8. The complete reduced two-spectator atlas

Exterior suspension strips off every spectator lying strictly outside the support interval. At $n = 7$, the irreducible two-spectator problem therefore consists of the six positive spacing words

$$113, \quad 122, \quad 131, \quad 212, \quad 221, \quad 311. \quad (8.1)$$

For each of the five generic words $\lambda \neq 212$, Note 23 transports one full-rank $n = 6$ edge anchor and proves that local compatibility completes it uniquely to an exact quotient coordinate

$$C_{\lambda} : E_{Q_{\lambda}} \longrightarrow T_7, \quad \ker C_{\lambda} = \text{im } \widehat{\partial}_{7, Q_{\lambda}}. \quad (8.2)$$

Consequently the full characteristic-zero dimension row is

λ	113	122	131	212	221	311
$\dim Y_{7, Q_\lambda}$	144	144	144	148	144	144.

(8.3)

The generic words fall into two edge-incidence classes. The exterior-type words 113, 311 have ranks

$$(144, 108, 144, 36, 108, 144), \quad (8.4)$$

whereas the three central words 122, 131, 221 have

$$(144, 108, 132, 36, 108, 144). \quad (8.5)$$

The latter lose precisely the $\mathbb{H} \otimes V$ spin profile

$$(1, 6, 5, 0, 0) \quad (8.6)$$

from the long-edge image. The exceptional word 212 instead restores the 144-dimensional standard core and adds the transverse cross-edge quaternion of Section 7. Hence the reduced layer has three exact structural classes:

113, 311 122, 131, 221 212	standard exterior-type profile central long-edge defect standard core plus cross-edge $\mathbb{H}$.
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(8.7)

In (8.6), the entries are the dimensions of the spin-0, 1, 2, 3, 4 isotypic subspaces, not the numbers of irreducible copies. They sum to 12.

Together with exterior suspension, this closes all fifteen tetrahedral supports at $n = 7$ over $\mathbb{Q}$. Quotient existence and dimension are no longer the finite problem. Note 24 takes the next step by comparing all transported direct-anchor completions.

9. The transported-anchor transition groupoid

Write the two ordered spectator factors as V_a, V_b , $a < b$, and put

$$T_7 = \mathbb{H}_L \otimes \mathbb{H}_R \otimes V_a \otimes V_b. \quad (9.1)$$

If $\phi_{a,r}$ applies an operator $\phi : \mathbb{H} \otimes V \rightarrow \mathbb{H} \otimes V$ to the indicated quaternion/vector pair, Note 24 proves that every transported direct-anchor transition is generated by

$A = (\theta^{-1})_{R,a},$	$B = (\theta^{-1})_{R,b},$	$S = (-\theta)_{L,b}.$
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(9.2)

The generic history chains are

122 : S, A,	131 : B, S, A,	221 : B, S,
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(9.3)

while all admissible histories collapse to one coordinate at 113 and 311. Across all six reduced words, twenty histories yield fourteen distinct core coordinates.

The factor supports determine the algebra:

$$AS = SA, \quad \text{rank}[A, B] = 64, \quad \text{rank}[S, B] = 51. \quad (9.4)$$

The full internal composite has

$$m_{ASB}(t) = (t-1)^2(t+1)^3(t^2+1)(t^2-t+1). \quad (9.5)$$

Thus overlapping copies of the first cross-product shear create Φ_4 and Φ_6 rotational factors. At the exceptional word the full transition is instead sharply block diagonal:

$$(C_1, \kappa_{212}) = (S \oplus I_{\mathbb{H}})(C_0, \kappa_{212}). \quad (9.6)$$

The extra quaternion is fixed rather than converted into a core shear. This rules out the simplest attempt to call the two-history comparison curvature. Note 25 then constructs the required transport between distinct spacing words and tests its only independent loop.

10. The spacing-word connection and localized defect

The six positive compositions of 5 form the reduced word graph

$$\Gamma_7 = \{113-122, 122-131, 122-212, 131-221, 212-221, 221-311\}. \quad (10.1)$$

An elementary edge slides one internal support vertex by one gap. The two support tetrahedra share a triangular face. Inside that face, the outer edge on the stationary side is present as the same decoded response variable in both local complexes.

For every adjacent pair λ, μ , the two selected 144-dimensional core coordinates have full rank and the same row space on that stationary hinge. Hence the equation

$$(C_\mu)_{h_{\lambda\mu}} = g_{\lambda\mu}(C_\lambda)_{h_{\lambda\mu}} \quad (10.2)$$

defines a unique integral $SO(3)$ -equivariant isomorphism. These six arrows are genuine inter-object transports, not coordinate changes on one quotient.

The graph has one independent cycle. Its exact holonomy is

$$g_{212,122}g_{221,212}g_{131,221}g_{122,131} = I_{144}. \quad (10.3)$$

Thus the complete core connection is flat. This does not make the gluing placement-blind. On the contrary, if all three common-face blocks are concatenated, the two trace row spaces are maximally transverse:

$$\dim(L_{\lambda,F} \cap L_{\mu,F}) = 0 \quad (10.4)$$

on every graph edge. Generic face trace spaces project isomorphically to their common 144-dimensional hinge row space. The full exceptional trace instead has dimension 148 and a four-dimensional projection kernel. Thus it is an extension over that hinge, not the graph of a function on it. These distinct face traces prevent naive facewise identification.

At the exceptional word,

$$Y_{7,Q_{212}} \cong T_{212} \oplus K_{212}, \quad K_{212} \cong \mathbb{H}. \quad (10.5)$$

Both graph hinges incident to 212 are outer edges, and the closed square coordinate vanishes on them:

$$(\kappa_{212})_{13} = (\kappa_{212})_{46} = 0. \quad (10.6)$$

Using the selected splitting, define the full maps to neighboring generic fibers by $\rho_{212,\mu} = g_{212,\mu} \pi_T$. These maps have kernel K_{212} . Their zero action on the residual is part of this definition: the hinge equation alone allows every extension

$$\rho_B(t, k) = g_{212,\mu} t + Bk, \quad B \in \text{Hom}(K_{212}, T_\mu).$$

Equivariance restricts B to equivariant maps; it does not force $B = 0$. For example, in the displayed T_7 model the nonzero injection $q \mapsto q \otimes 1 \otimes \sum_a e_a \otimes e_a$ is equivariant. Thus the intrinsic assertion is invisibility on the hinges; forgetting is the chosen extension. With that convention the complete finite answer is

$$\boxed{\text{flat transportable 144-core} + 212\text{-supported quaternionic defect.}} \quad (10.7)$$

The core loop is exactly flat and the defect does not enter the chosen hinge transport. This separates chart shear, word holonomy, and placement residual within the finite construction.

11. Three internal spectators: the 222 residual and descent obstruction

We now use actual gap labels. Let $Q_{222} = (1, 3, 5, 7)$, with spectators 2, 4, 6. Removing these spectators one at a time gives parents 122, 212, and 221, respectively. The combinatorial deletion alone does not define a map of quotients.

11.1 The intrinsic residual quotient

For a five-probe response define

$$\epsilon_5(F)(z) = \sum_{a,b=1}^3 F(e_a, e_a, z, e_b, e_b),$$

and put

$$\kappa_{222}(F) = \epsilon_5(F_{15}) - \epsilon_5(F_{17}) - \epsilon_5(F_{35}) + \epsilon_5(F_{37}).$$

The blocks on 13 and 57 are zero. For $v \in V$, the identity $\sum_a e_a v e_a = v$ removes the two paired probes. Each literal cross-edge response reduces to the same response

$$v_8 v_7 v_6 v_5 z v_4 v_3 v_2 v_1.$$

Surjectivity of each terminal face therefore makes its two cross-edge compositions agree. Their signed incidence contributions cancel, giving $\kappa_{222} \partial_{222} = 0$. The three independent choices of the open probe z , each with four output components, show that the map is onto $\mathcal{R}_1$.

Theorem 11.1 (the 222 object). Over $\mathbb{Q}$,

$$\text{rank } \partial_{222} = 5388, \quad \dim Y_{222} = 444.$$

The dimensions of the labelled edge images are

Actual edge	13	15	17	35	37	57
Image dimension	432	336	408	120	336	432

The two outer-edge images coincide and give an intrinsic exact sequence

$$0 \longrightarrow T_{222} \longrightarrow Y_{222} \xrightarrow{\bar{\kappa}_{222}} \mathcal{R}_1 \longrightarrow 0, \quad \dim T_{222} = 432.$$

In particular,

$$T_{222} = E_{13}^Y = E_{57}^Y = \ker \bar{\kappa}_{222}, \quad D_{222} := Y_{222}/T_{222} \cong \mathcal{R}_1.$$

No complementary 12-dimensional subspace inside Y_{222} is chosen.

Proof. The companion certificate constructs the rational matching in natural face coordinates and right-decoded edge coordinates. Its target dimension is 5832. A reconstructed 444-row annihilator is checked against the matching over the integers and has full row rank. Matching ranks 5388 over both prime fields give the opposite bound over $\mathbb{Q}$. The six edge ranks and the 432-dimensional joint outer-edge image are verified using exact rational row identities. Since κ_{222} is onto and vanishes on the outer edges, each 432-dimensional outer image equals its kernel in Y_{222} . This proves the sequence and the equalities. Full construction and all finite matrices are reproducible from Note 26's certificate.

On the generic parent sides, Note 22's exterior edge suspension gives

$$Y_{122} \otimes V \xrightarrow{\sim} T_{222}, \quad Y_{221} \otimes V \xrightarrow{\sim} T_{222}.$$

The first uses actual child edge 13 and the second uses 57 (with the standard tensor flip for the left suspension). In each case the carried parent block and the child block have the same kernel and rank 432. This proves the unique descended isomorphism onto the core; it does not define a projection from the complete child quotient onto that core.

11.2 A specified central-slot decoder comparison

For the central parent, put $D_{212} = Y_{212}/T_{212}$, where $T_{212} = \ker \bar{\kappa}_{212}$ is the common 144-dimensional outer-edge image. Thus $D_{212} \cong \mathbb{H}$. The anchored summand K_{212} of Section 7 maps isomorphically onto this intrinsic quotient.

Choose the **central-slot right decoder suspension**. On every edge, insert an argument z at actual gap 4, but multiply its value on the right:

$$(\Sigma_4^R(F \otimes w))(\dots, z, \dots) = F(\dots, \hat{z}, \dots) z w.$$

This is an explicitly chosen response operator. It is not identified with inserting a vector into a literal state word, and no full chain-map property is assumed. Invertibility of the local encoder makes it an isomorphism $E_{212} \otimes V \rightarrow E_{222}$.

On the cross edges, the open argument is the middle one, so the paired sums give the exact residual square

$$\boxed{\kappa_{222} \Sigma_4^R = \Theta_R(\kappa_{212} \otimes \text{id}_V), \quad \Theta_R(k \otimes w)(z) = k z w.}$$

This identity is checked edge by edge in the certificate as well.

Theorem 11.2 (residual comparison and full descent obstruction). The chosen suspension induces an isomorphism

$$\beta_R : D_{212} \otimes V \xrightarrow{\sim} D_{222}, \quad \text{rank } \beta_R = 12, \quad \ker \beta_R = 0.$$

For a fixed unit $w \in V$, the corresponding map on the parent residual has rank four and zero kernel, with readout

$$f_k(z) = kzw, \quad k = -f_k(w).$$

However, the full suspension does not descend to $Y_{212} \otimes V \rightarrow Y_{222}$. Its obstruction

$$\mathcal{O} = \pi_{222} \Sigma_4^R(\partial_{212} \otimes \text{id}_V)$$

satisfies

$$\boxed{\text{im } \mathcal{O} = T_{222}, \quad \text{rank } \mathcal{O} = 432.}$$

Hence D_{222} is exactly the largest quotient of Y_{222} on which this suspension descends from $Y_{212} \otimes V$.

Proof. The residual square sends old matching relations into $\ker \bar{\kappa}_{222} = T_{222}$. Passing to that quotient gives $\Theta_R(\bar{\kappa}_{212} \otimes \text{id}_V)$; its kernel is $T_{212} \otimes V$ and its induced residual map is an isomorphism. For fixed unit w , $w^2 = -1$ gives the displayed inverse readout. The same residual square bounds the obstruction rank by 432. The certificate exhibits rank 432 modulo each prime, using operators whose exact rational identities have been verified. The upper and lower bounds agree. Every quotient admitting full descent must kill this entire image; quotienting by it is also sufficient. This proves the final universal property.

11.3 What has and has not been transported

The earlier projection-defined maps forget K_{212} . The operation in this section is a different, specified comparison into a new residual quotient; its successful readout does not reverse those information-losing hinges. The right decoder choice is part of Theorem 11.2. A choice-independent full transport, or a local rule that turns the representative ambiguity into a new core state, is not provided.

The finite obstruction supplies a concrete next question: which additional local condition, if any, determines a representative or a correction? It is not a time-update law and is not nonzero core holonomy.

12. Results and scope

Stripped of the surrounding depth-filtration narrative, Notes 14–26 give the following self-contained package.

1. **Closed tetrahedral quotient.** The minimal six-edge relation object is explicitly $\mathbb{H} \otimes \mathbb{H}$, with a canonical $12 + 4$ orthogonal channel split.
2. **Placement-sensitive decoration.** Isomorphic total quotients can be distinguished by the incidence of their six labelled edge images.
3. **Two-chart spectator atlas.** The five one-spectator placements require two direct seed charts, and the existence and nonexistence of every chart anchor are exact over $\mathbb{Q}$.
4. **Central cross-product transition.** The exterior overlap is the identity, while the central overlap is the explicit reflection-plus-shear map G .
5. **Channel transfer.** The central placement changes which orthogonal seed channel carries the surviving quaternion.
6. **Invariant residual.** The coordinate decoder changes across the overlap, but the quaternion-valued cap residual does not.
7. **Parity continuation.** The first admissible two-spectator cross square produces an exact quaternionic residual at $n = 7$.

8. **Exceptional core–residual decomposition.** A transported spectator core and the cross square give the exact split $144 + 4$.
9. **Exterior functoriality.** Left and right exterior spectators tensorize every quotient and labelled edge image, and the two suspensions commute.
10. **Complete reduced two-spectator layer.** All six internal words have exact rational coordinates and certified edge-incidence profiles.
11. **Three-generator transition groupoid.** All transported-anchor histories are generated by pair-local A, B, S , and the exceptional residual is fixed by its full transition.
12. **Stationary-edge word connection.** Every adjacent reduced word pair has a unique integral equivariant core transport descending from its shared outer hinge, and the unique independent loop is exactly flat.
13. **Localized quaternionic defect.** Full common-face traces are transverse, while the exceptional quaternion vanishes on both incident hinges and is supported only at 212.
14. **Three-spectator residual comparison.** The word 222 has an intrinsic 432-dimensional outer core and a 12-dimensional residual quotient. A specified right decoder induces an isomorphism from $D_{212} \otimes V$ onto that quotient, while its full descent obstruction fills the core.

Ordered placement determines which local charts exist. Their transitions define the stationary-edge core connection, while the exceptional residual is invisible on the two incident hinges.

This is a complete finite algebraic atlas through two spectators, together with its exterior towers, same-word transition groupoid, adjacent-word connection, and closed-loop test. Calling the quaternion curvature would still be premature: the first independently defined loop is flat, and the quaternion lies in the kernel of both selected projection-defined maps. Section 11 settles one three-internal-spectator object and a specified residual comparison. The remaining three-spectator word complex and any residual-coupled holonomy remain open.

13. Sources and reproducibility

Note	Role in this synthesis
14	closed seed operator ω_5 and the first $12 + 4$ split
15	decorated edge images and placement classification
16	all-odd parity square and exact $n = 7$ cross residual
17	all-even cap complex and exact $n = 6$ residual
18	Frobenius coordinate and orthogonal seed channels
19	normalized $n = 6$ quotient, contraction β , and channel transfer
20	complete one-spectator atlas and central transition
21	exact exceptional $144 + 4$ core-residual coordinate
22	all-length exterior suspension and strict left/right flatness
23	complete rational atlas of the six reduced $n = 7$ words
24	exhaustive anchor histories, three transition generators, and exceptional residual invariance
25	stationary-edge word transport, core flatness, face transversality, and localized quaternionic defect
26	444-dimensional 222 quotient, residual comparison, and full descent obstruction

The companion archive supplies the original notes and the following certificates. Earlier certificates retain their original claim boundaries; the release verification record identifies which were executed for this snapshot.

`certificates/n8_222_redetection_certificate.py`

The earlier exact computations are supplied by

`certificates/n5_quaternionic_second_differential_certificate.py`
`certificates/n5_central_channel_factorization_certificate.py`
`certificates/spectator_placement_residual_certificate.py`
`certificates/n7_exceptional_square_operator_certificate.py`
`certificates/n6_capped_five_edge_operator_certificate.py`
`certificates/n6_seed_cap_bridge_certificate.py`
`certificates/n6_spectator_chart_transition_certificate.py`
`certificates/n7_exceptional_core_decomposition_certificate.py`
`certificates/exterior_spectator_suspension_certificate.py`
`certificates/n7_internal_word_atlas_certificate.py`
`certificates/n7_anchor_transition_groupoid_certificate.py`
`certificates/n7_spacing_word_transport_certificate.py`

14. Provenance and mathematical dependencies

Section 1 gives the local definitions. Complete derivations and computational proof components are retained in the companion notes and certificates.

Foundation: Residual Chart Lab, *Free Numbers Core v1.0.0* (2026), DOI [10.5281/zenodo.21328471](https://doi.org/10.5281/zenodo.21328471).

Source repository: baseline 09b77f329d0c825c794443a0cd98cde89d8f1d73; Note 26 commit 881ff66. The claim ledger, verification records, manifest and checksums fix the scope and reproducibility of this MIT-licensed snapshot. No literature-wide novelty or physical interpretation is claimed.